

Java Track, lesson 1.1e

Printing data

Java gives you three ways to put text on the console: println adds a line break, print does not, and printf builds the line from a template with format specifiers.

Three ways to print

print

Prints and keeps the cursor on the same line.

```
System.out.print("Hi, ");
```

println

Prints, then moves to a new line.

```
System.out.println("Greg");
```

printf

Fills a template with values.

```
System.out.printf(
  "%s: %d%n", "Lessons", 5);
```

print vs println

```
System.out.print("A");
System.out.println("B");
System.out.print("C");
```

Console:

```
AB
C
```

println() with no argument prints only an empty line.

Format specifiers

`%s`	String, or any object
`%d`	Whole number: int, long
`%f`	Decimal number: double
`%.2f`	Decimal, 2 digits after the separator
`%5d`	Width of 5 characters
`%n`	Line break for this operating system
`%%`	A literal percent sign

Practical example

```
String name = "Greg";
int lessons = 5;
double score = 8.456;

System.out.printf("%s finished %d lessons%n", name, lessons);
System.out.printf("Average score: %.2f%n", score);  // 8.46
System.out.printf("Progress: %d%%%n", 40);           // 40%
String line = "Score: %.1f".formatted(score);    // Score: 8.5
```

Formatting without printing

```
String a = String.format("%.1f", 8.456);   // "8.5"
String b = "%s scored %d".formatted("Ana", 9); // "Ana scored 9"
```

Same specifiers as printf. You will meet them again in log messages and reports.

Common mistakes

- Beginners expect "Total: " + 1 + 2 to print Total: 3. It prints Total: 12, because once a String is involved, + joins text. Use "Total: " + (1 + 2).
- Beginners forget that printf adds no line break. Without %n the next output ends up on the same line.
- Beginners use the wrong specifier, such as %d with a double. That compiles but throws an IllegalFormatConversionException at runtime.
- printf uses the computer's locale: on a Dutch system %.2f prints 8,46. Use printf(Locale.US, ...) when you need a dot.

The Dutch comma

```
System.out.printf("%.2f%n", 8.456);              // 8,46 on a Dutch Mac
System.out.printf(Locale.US, "%.2f%n", 8.456);  // 8.46 everywhere
```

printf, String.format and formatted use the default locale unless you pass one.

New in Java 25: IO

```
void main() {
    IO.println("Hello");
    IO.print("No line break");
}
```

A compact source file with an instance main method is a complete program. IO lives in java.lang, so no import. System.out still works everywhere and is what you will see in existing code.

Remember

Java gives you three ways to put text on the console: println adds a line break, print does not, and printf builds the line from a template with format specifiers.

Next: basic terms JVM, JRE and JDK.